MathsJam Shout

~July 2025
Approximation Special

Since this month's second-to-last Tuesday is 22/7 (a great approximation to π), we've made this month's Shout an homage to the concept of approximation in various forms. Hope it's approximately loads of fun!

Calculate Continued Fractions

You can use continued fractions to get closer and closer approximations to irrational numbers like square roots and e.

$$\pi = 3 + \cfrac{1}{7 + \cfrac{1}{15 + \cfrac{1}{1 + \frac{1}{292} + \cdots}}}$$

Find the continued fractions for various interesting constants, and see how close you can get!

en.wikipedia.org/wiki/Simple_continued_fraction

Estimate Fermi Questions

Fermi estimation is a method for finding approximate solutions to questions it would be impractical or impossible to measure the answer to precisely (like, how many pints of beer are drunk in the UK every day?) Search online for examples, or come up with your own!

Try It Rectangle Approximation

Christian Lawson-Perfect, former organiser of Newcastle MathsJam, came up with a method of approximating the ratio of sides on a rectangle - you could try this with a napkin or piece of paper in the pub!

Instructions and more information here:

bit.ly/ratio-folding

Follow It Approx Bot

Matthew Scroggs, the organiser of King's Lynn MathsJam, has built a bot which expresses the date as a mathematically interesting number it's a good approximation to. What's your birthday a good approximation to? Scroll back in the feed to find out!

mathstodon.xyz/@HappyApproxDay

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com** for details.

MathsJam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly MathsJam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest MathsJam, visit the website at **www.mathsjam.com**.